



COMSATS University Islamabad

Department of Computer Science

Course Description Form (CDF)

Course Information

Course Code & Title: **CSC101 - Applications of Information and Communication Technologies**

Credit Hours: **3 (2, 1)**

Lecture Hours/Week: **2**

Lab Hours/Week: **3**

Pre-Requisites: **None**

Course objective:

- To foster a comprehensive understanding of ICT concepts, including computing models and components;
- To navigate databases, networks, and information systems for effective communication;
- To explore emerging technologies like cloud computing and artificial intelligence;
- To explore real-world ICT applications in domains such as e-commerce and security;
- To gain practical problem-solving skills through hands-on exercises and gain insights from case studies of ICT applications in various domains.

Course Contents:

This course covers the basics of Information and Communications Technologies: Topics include: Overview of ICT; Computing Models; Computing Devices & their Types; Computer Systems & Components; Number Systems & Computer Codes; System & Application Software; Introduction to Databases & Information Systems; Computer Networks & Internet; E-Commerce; Security & Privacy; Intellectual Property Rights; ICT Applications: Health Industry, Banking, Expert Systems, and Digital Currencies; Emerging Trends in ICT: Cloud Computing, IoT, Virtual & Augmented Reality, Uses of Artificial Intelligence; and Problem Solving using Computer Programming.

Unit wise Major Topics:

Unit	Topic	No. of Teaching Hours
1.	ICT & Computing Overview: ICT Introduction; Computing Models; Computing Devices, Types & Components;	3
2	Number Systems: Binary, Octal, Hexadecimal, and their Conversions; Binary Arithmetic; and Computer Codes.	3
3.	ICT Domain: System Software: Operating Systems, Application Software; Introduction to Databases & Information Systems; Computer Networks; E-commerce; Security & Privacy; and Intellectual Property Rights;	9
4.	ICT Applications: Health Industry, Banking, Expert Systems, and Digital Currencies; Emerging Trends: Cloud Computing, IoT, and Augmented Reality; and Uses of Artificial Intelligence.	5
5.	Problem Solving using Computer Programming: Concepts, Types, process; Algorithm Representation;	4
6	Program Development Lifecycle; Programming Language (Python): Data Types; Variables & Expression; Elementary Programming; Control Structures; Loops; and Functions.	6

Total Contact Hours						30	
Mapping of CLOs and GAs							
Sr.#	Unit #	Course Learning Outcomes				Blooms Taxonomy Learning Level	GA
CLO's for Theory							
CLO-1	1	Explain the basic computing models and related hardware.				Understanding	2
CLO-2	2	Work out with different number systems and codes.				Applying	2,3
CLO-3	3-4	Describe the fundamental concepts of ICT domains.				Understanding	2
CLO-4	5	Solve computing problems using problem-solving process.				Applying	2,3,5
CLO-5	6	Describe the concepts of variables, conditional, repetitive structures, and functions using a programing language.				Understanding	2,3
CLO's for Lab							
CLO-6	3-5	Demonstrate the usage of productivity software, problem solving tool, and web technology by performing appropriate tasks.				Applying	5
CLO-7	6	Write a program in a programming language to solve a simple problem.				Applying	2-5
CLO Assessment Mechanism							
Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5	CLO-6	CLO-7
Quizzes	Quiz 1	Quiz 2	Quiz 3	Quiz 4	-	-	-
Assignments	-	-	Assignment 1&2	Assignment 3	Assignment 4	Lab Assignment 1-3	Lab Assignment 4
Mid Term Exam	Mid Term Exam	Mid Term Exam	Mid Term Exam	-	-	Lab Mid Term Exam	-
Final Term Exam	Final Term Exam					-	Lab Project/ Lab Final Term Exam
Text and Reference Books							
Textbooks:							
1. Computer Science Illuminated, Nell Dale and John Lewis, Jones & Bartlett Learning, LLC, 2020							
2. Understanding Computers: Today and Tomorrow, Comprehensive, Deborah Morley, Charles S. Parker, Cengage Learning, 2017.							
3. Python Basics: A Practical Introductio							

4. Python Programming an Introduction to Computer Science, John M. Zelle, Franklin, Beedle & Associates Inc, 2016.

Reference Books:

5. Foundations of Computer Science, Behrouz Forouzan, McGraw-Hill, 2017.
6. Problem Solving & Programming, Maureen Sprankle, Jim Hubbard, Prentice Hall, 2012.